

Pooja Thalur

6023804300 | thalur.pooja15@gmail.com | [linkedin.com/in/poojathalur](https://www.linkedin.com/in/poojathalur) | Tempe, AZ 85281

SUMMARY:

- Software Engineer with over 5 years of experience in backend development, cloud-native architectures, and scalable web applications.
- Proficient in Java, Spring Boot, Python, and Grails, with hands-on expertise in microservices, REST APIs, and distributed systems.
- Extensive experience working with cloud platforms like AWS and Azure for storage, compute, and large-scale migrations.
- Built and optimized high-performance notification, scheduling, and payment systems using Quartz, Razorpay, and batch processing.
- Adept at integrating third-party APIs (Zoom, Twilio, Meta, Exotel) and implementing secure auth mechanisms (OAuth2, JWT).
- Designed GDPR-compliant features for data retention, cookie consent, and user privacy across SEO-focused React applications.
- Led containerization and CI/CD efforts using Docker, Jenkins, and Git to streamline deployment and reduce release failures.
- Skilled in monitoring and observability using ELK stack, New Relic, and custom dashboards for proactive issue resolution.
- Experience with A/B testing, SEO optimization, referral systems, and revenue engines to drive growth and engagement.
- Collaborated across product, QA, and dev teams in Agile environments to deliver robust features across multiple sprints.
- Strong command of data analysis tools including Pandas, Databricks, Tableau, and Matplotlib for insights and reporting.
- Passionate mentor and contributor to code reviews, documentation, and continuous improvement initiatives.

TECHNICAL SKILLS:

Languages	Java, Python, Groovy, C++, Typescript, JavaScript, HTML, CSS, SQL
Web Frameworks	Spring Boot, MVC, Grails, Django
Front-End Technologies	React.js, HTML/CSS, JavaScript, Typescript
Databases	MySQL, PostgreSQL, MongoDB, Memcached
Web Technologies	jQuery, AJAX, Requests, re, RESTful APIs
Messaging & Search	ActiveMQ, RabbitMQ, Kafka, Solr
Cloud Platforms	AWS (EC2, S3, CloudFront, Lambda), Azure
Containerization/CI/CD	Docker, Jenkins, Git, CI/CD, Tomcat
Big Data & ML	Spark, TensorFlow, PyTorch, NumPy, Pandas, Matplotlib, OpenAI API, Model Evaluation
Testing Tools	JUnit, Mockito, Selenium, Postman
Web Technologies	RESTful APIs
Project Management	Jira, Confluence

EDUCATION:

- M.S., Information Technology, Arizona State University, Tempe, AZ, May 2025
- B.E., Computer Science, Visvesvaraya Technological University, Bengaluru, India, July 2019

PROFESSIONAL EXPERIENCE:

Software Developer: Digital Innovation Group, ASU

March 2024 – May

2025

- Designed and developed scalable microservices in Spring Boot for processing real-time bibliographic metadata.
- Upgraded legacy Java and Neo4j systems to improve security, performance, and maintainability in a distributed setup.
- Created asynchronous APIs and comprehensive unit tests, increasing code coverage and reducing regression bugs by 30%.
- Built and consumed RESTful APIs in Java and Python for internal and third-party integrations.
- Dockerized microservices, cutting deployment time and environment-specific issues by 25%.
- Participated in peer code reviews and Agile ceremonies for continuous improvement.
- Led container orchestration and troubleshooting in Linux-based environments.
- Streamlined application logs and monitoring pipelines with ELK stack for faster root-cause analysis.
- Collaborated with cross-functional research teams to convert legacy solutions into interactive digital exhibits.
- Integrated secure authentication layers in internal APIs for data-sensitive research systems.
- Worked with Git, Jenkins, and Maven for version control and automated deployments.
- Provided end-user training and documentation for internal digital archiving tools.

Environment: Java, Python, Spring Boot, REST APIs, Neo4j, Docker, Jenkins, Git, Maven, Linux, ELK Stack, OAuth, Agile.

Software Engineer II: UrbanPro.com

April 2023 – August 2023

- Developed batch notification system for automated marketing via email, SMS, and app, reducing manual tasks by 50%.
- Designed a Money Management System integrating RazorPay for tutor payouts and New Relic for monitoring performance.
- Created SEO-friendly tutor profiles using Quartz jobs and CDNs, improving page generation time by 50%.
- Built scheduling automation for 1-hour Master Classes, increasing student enrollments by 15%.
- Migrated over 10TB of media assets from AWS to Azure Blob Storage with minimal downtime.
- Switched WhatsApp messaging provider from Twilio to Meta, saving over \$50,000 in yearly costs.
- Refactored background workers and job queues for improved fault tolerance and performance.
- Enhanced system observability with centralized dashboards and real-time alerts.
- Spearheaded GDPR-compliant data retention and notification strategies across microservices.
- Collaborated with product and QA teams to streamline release cycles with CI/CD pipelines.
- Delivered feature enhancements with zero rollback across 5 agile sprints.
- Mentored junior developers and reviewed PRs for best practices in code quality.

Environment: Java, Spring Boot, Razorpay, AWS, Azure Blob Storage, Twilio, Meta APIs, Quartz, New Relic, Docker, CI/CD, Git, Microservices, Agile.

Software Engineer: UrbanPro.com

April 2021 – March 2023

- Engineered an ad-driven revenue engine for tutors using Spring Boot and Memcached, increasing ROI by 20%.
- Integrated Zoom OAuth and JWT-based video call services for secure tutor-student interactions.
- Built scalable referral system with tracking for students and tutors, improving conversion by 15%.
- Migrated legacy webhooks (Zoom, Exotel) to OAuth-secured integrations, enhancing audit logging.
- Implemented GDPR-compliant cookie consent and deletion workflows across SEO-heavy React pages.
- Created A/B test frameworks in UI to optimize pricing models, increasing conversions by 15%.
- Developed policy violation reporting dashboard, reducing misuse by 30%.
- Refactored user ranking logic and search caching for category-level tutor discovery.
- Authored microservices for progress tracking and feedback loops in the learning dashboard.
- Deployed and monitored apps via Jenkins CI pipelines and Docker containers in production.
- Assisted in implementing advanced access controls and user privacy settings site-wide.
- Managed backend error tracking and logging strategies to reduce MTTR.

Environment: Java, Spring Boot, React, Memcached, Zoom OAuth, JWT, Exotel, Docker, Jenkins, OAuth2, A/B Testing, Microservices, Git, Agile.

Associate Software Engineer: UrbanPro.com

October 2019 - March 2021

- Developed demo class booking system using Grails, improving tutor visibility and booking rates by 30%.
- Implemented strategy patterns for dynamic SEO page generation for seasonal classes.
- Integrated Exotel API to mask tutor-student calls, reducing revenue leakage by 30%.
- Revamped tutor ranking logic to reduce customer support complaints by 20%.
- Refactored core database queries for optimized class scheduling under high load.
- Wrote detailed documentation for onboarding and product usage.
- Participated in bi-weekly sprints and demos with stakeholders.
- Created REST endpoints to serve hobby class data in mobile apps.
- Set up Jenkins jobs for nightly builds and regression testing.
- Introduced centralized configuration service to ease environment setup across teams.
- Supported QA team with post-deployment validations and fixes.
- Collaborated in backend refactoring to support new business use cases.

Environment: Grails, Groovy, Java, Exotel API, REST APIs, Jenkins, PostgreSQL, Git, Agile, Strategy Pattern, Microservices.

Data Science Intern: Yugasys

May 2018 - July 2018

- Built a web-based dashboard using Angular and REST APIs for application health monitoring.
- Performed web scraping and data cleaning in Python for large-scale datasets.
- Utilized Databricks for scalable ETL jobs and processing performance metrics.
- Designed interactive dashboards with D3.js and Tableau for actionable insights.
- Conducted exploratory data analysis using Pandas and Matplotlib.
- Worked on user behavior modeling for app optimization.
- Built basic ML classification models for usage prediction.
- Collaborated with senior engineers to integrate visualizations into client reporting tools.
- Automated periodic data pulls using cron jobs.
- Created user documentation for internal analytics workflows.
- Attended agile standups and presented findings to product managers.
- Implemented pagination and filtering in front-end dashboard.

Environment: Python, Angular, REST APIs, Databricks, Pandas, Matplotlib, D3.js, Tableau, Cron, ML (Scikit-learn), HTML/CSS, Agile.

Projects:

Predictive Modeling for Credit Card Default Prediction

Built robust ML models for credit card default prediction. Used data processing techniques, feature engineering, and identified Random forest as best with 82% accuracy and high F1-score.

Scalable Image Recognition System with AWS

High performance image recognition system using AWS EC2, SQS, S3 and Lambda, optimizing auto-scaling to dynamically allocate resources to process 1000 images in 50 seconds with up to 20 instances for enhanced performance.